

HW1

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1 Problem 1.1

The problem defined was to find the minimum of the function

$$f(x_1, x_2) = (x_1 - 1)^2 + 2(x_2 - 2)^2, \quad (1)$$

subject to the constraint

$$g(x_1, x_2) = x_1^2 + x_2^2 - 1 \leq 0. \quad (2)$$

The method used for solving the problem is the penalty method. The penalty method is a method utilizing a function which transforms a constrained problem to an unconstrained one. The goal is to minimize the sum of the penalty function, which depends on the constraints and the objective function. The general penalty function is the following:

$$p(\mathbf{x}; \mu) = \mu \left(\sum_{i=1}^m (\max\{g_i(\mathbf{x}), 0\})^2 + \sum_{i=1}^k (h_i(\mathbf{x}))^2 \right) \quad (3)$$

Where g_i represents an inequality constraint, h_i an equality constraint, and $\mu > 0$ the penalty parameter.

Minimizing f subject to the constraint g corresponds to minimizing the function $f_p(\mathbf{x}; \mu)$:

$$f_p(x_1, x_2; \mu) = f(x_1, x_2) + \mu (\max\{g(x_1, x_2), 0\})^2 \quad (4)$$

Substituting $f(x_1, x_2) = (x_1 - 1)^2 + 2(x_2 - 2)^2$ and $g(x_1, x_2) = x_1^2 + x_2^2 - 1$ into eq (4) :

$$f_p(x_1, x_2; \mu) = (x_1 - 1)^2 + 2(x_2 - 2)^2 + \mu (\max\{x_1^2 + x_2^2 - 1, 0\})^2 \quad (5)$$

2. To calculate the gradient $\nabla f_p(\mathbf{x}; \mu)$, differentiate $f_p(x_1, x_2; \mu)$ with respect to x_1 and x_2 .

Evaluate the gradient for the two different cases:

Case 1: Constraint is fulfilled ($x_1^2 + x_2^2 - 1 \leq 0$)

Inside or on the boundary of the constraint, $\max\{x_1^2 + x_2^2 - 1, 0\} = 0$. The penalty term is 0. Therefore, the gradient is:

$$\nabla f_p(\mathbf{x}; \mu) = \begin{bmatrix} 2(x_1 - 1) \\ 4(x_2 - 2) \end{bmatrix} \quad (6)$$

Case 2: Constraint is violated ($x_1^2 + x_2^2 - 1 > 0$)

Outside the constraint, $\max\{x_1^2 + x_2^2 - 1, 0\} = x_1^2 + x_2^2 - 1$. The function becomes:

$$f_p(x_1, x_2; \mu) = (x_1 - 1)^2 + 2(x_2 - 2)^2 + \mu(x_1^2 + x_2^2 - 1)^2 \quad (7)$$

Applying the chain rule to each component:

$$\begin{aligned} \frac{\partial f_p}{\partial x_1} &= 2(x_1 - 1) + 2\mu(x_1^2 + x_2^2 - 1) \cdot \frac{\partial}{\partial x_1}(x_1^2 + x_2^2 - 1) \\ &= 2(x_1 - 1) + 4\mu x_1(x_1^2 + x_2^2 - 1) \end{aligned} \quad (8)$$

$$\begin{aligned} \frac{\partial f_p}{\partial x_2} &= 4(x_2 - 2) + 2\mu(x_1^2 + x_2^2 - 1) \cdot \frac{\partial}{\partial x_2}(x_1^2 + x_2^2 - 1) \\ &= 4(x_2 - 2) + 4\mu x_2(x_1^2 + x_2^2 - 1) \end{aligned} \quad (9)$$

Therefore, the gradient is:

$$\nabla f_p(\mathbf{x}; \mu) = \begin{bmatrix} 2(x_1 - 1) + 4\mu x_1(x_1^2 + x_2^2 - 1) \\ 4(x_2 - 2) + 4\mu x_2(x_1^2 + x_2^2 - 1) \end{bmatrix} \quad (10)$$

Summary

$$\nabla f_p(\mathbf{x}; \mu) = \begin{cases} \begin{bmatrix} 2(x_1 - 1) \\ 4(x_2 - 2) \end{bmatrix}, & \text{if } x_1^2 + x_2^2 - 1 \leq 0 \\ \begin{bmatrix} 2(x_1 - 1) + 4\mu x_1(x_1^2 + x_2^2 - 1) \\ 4(x_2 - 2) + 4\mu x_2(x_1^2 + x_2^2 - 1) \end{bmatrix}, & \text{if } x_1^2 + x_2^2 - 1 > 0 \end{cases} \quad (11)$$

The minimum of the objective function $f(x_1, x_2)$ without taking the constraint into account is calculated by finding the stationary point where the gradient is zero. $\nabla f(\mathbf{x}) = \mathbf{0}$:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2(x_1 - 1) \\ 4(x_2 - 2) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (12)$$

Which gives:

$$2(x_1 - 1) = 0 \implies x_1 = 1 \quad (13)$$

$$4(x_2 - 2) = 0 \implies x_2 = 2 \quad (14)$$

Therefore, the unconstrained minimum is at $(x_1, x_2) = (1, 2)$.

The gradient descent algorithm was done with the step length $\eta = 0.0001$, a stopping tolerance of $T = 10^{-6}$, and the penalty parameters $\mu \in \{1, 10, 100, 1000\}$. The $\mathbf{x}$ -values $\mathbf{x}^*(\mu)$ and the constraint values $g(\mathbf{x}^*)$ calculated are found in Table 1.

Table 1: $f_p(\mathbf{x}; \mu)$ for varying penalty parameters μ .

μ	x_1^*	x_2^*
1.0	0.4338	1.2102
10.0	0.3314	0.9955
100.0	0.3137	0.9553
1000.0	0.3118	0.9507

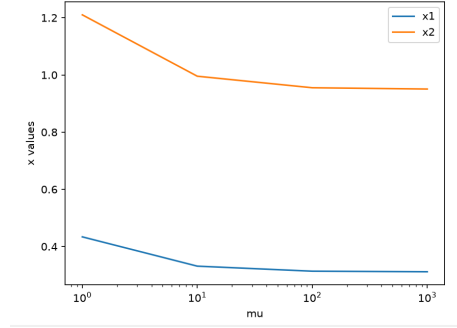


Figure 1: Convergence Plot

As one can observe in the Table 1 and the convergence plot the sequence of points $\mathbf{x}^*(\mu)$ converges towards the constraint boundary. At $\mu = 1000$, the point is $\mathbf{x}^* \approx (0.3118, 0.9507)$, which gives $x_1^2 + x_2^2 - 1 \approx 0.3118^2 + 0.9507^2 - 1 \approx 0.0010 \approx 0$. That can be seen to confirm that the sequence approximates the constrained minimum on the unit circle $x_1^2 + x_2^2 = 1$.

2 Problem 1.2

The problem is finding the maximum and minimum values (and their respective points) of the function

$$f(x_1, x_2) = 4x_1^2 + 2x_2^3$$

over the closed set

$$S = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 \leq 4\}.$$

Evaluate:

1. Critical points in the interior of S ($x_1^2 + x_2^2 < 4$).
2. Critical points on the boundary of S ($x_1^2 + x_2^2 = 4$) in this case with the Lagrange multiplier method.

1. Critical Points in the interior

In the interior of S candidate points for local extrema are stationary points where the gradient of f equals 0.

$$\nabla f(x_1, x_2) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 8x_1 \\ 6x_2^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This gives:

$$8x_1 = 0 \implies x_1 = 0,$$

$$6x_2^2 = 0 \implies x_2 = 0.$$

The only critical point in the interior is $(0, 0)$. $0^2 + 0^2 = 0 < 4$ so the point lies strictly on the interior S . The function value at the point is:

$$f(0, 0) = 4(0)^2 + 2(0)^3 = 0.$$

2. Critical Points on the boundary

The boundary is defined by the equality constraint:

$$h(x_1, x_2) = x_1^2 + x_2^2 - 4 = 0.$$

The function used in the Lagrange method:

$$\mathcal{L}(x_1, x_2, \lambda) = f(x_1, x_2) + \lambda h(x_1, x_2) = 4x_1^2 + 2x_2^3 + \lambda(x_1^2 + x_2^2 - 4).$$

Setting the gradient of $\mathcal{L}$ to zero ($\nabla \mathcal{L} = \mathbf{0}$) gives the following system of equations:

$$\frac{\partial \mathcal{L}}{\partial x_1} = 8x_1 + 2\lambda x_1 = 2x_1(4 + \lambda) = 0 \tag{15}$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = 6x_2^2 + 2\lambda x_2 = 2x_2(3x_2 + \lambda) = 0 \tag{16}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = x_1^2 + x_2^2 - 4 = 0 \tag{17}$$

From equation (15), we obtain two cases: $x_1 = 0$ or $\lambda = -4$.

Case 1: $x_1 = 0$ Substituting $x_1 = 0$ into the boundary constraint (17):

$$0^2 + x_2^2 = 4 \implies x_2 = \pm 2.$$

- For $x_2 = 2$: From (16), $2(2)(3(2) + \lambda) = 0 \implies \lambda = -6$.

$$\text{Point A: } (0, 2), \quad f(0, 2) = 4(0)^2 + 2(2)^3 = 16.$$

- For $x_2 = -2$: From (16), $2(-2)(3(-2) + \lambda) = 0 \implies \lambda = 6$.

$$\text{Point B: } (0, -2), \quad f(0, -2) = 4(0)^2 + 2(-2)^3 = -16.$$

Case 2: $\lambda = -4$ ($x_1 \neq 0$) Substituting $\lambda = -4$ into equation (16):

$$2x_2(3x_2 - 4) = 0.$$

This produces two subcases for x_2 :

- **Subcase 2a:** $x_2 = 0$

Substituting $x_2 = 0$ into the constraint:

$$x_1^2 + 0^2 = 4 \implies x_1 = \pm 2.$$

This gives two points:

$$\text{Point C: } (2, 0), \quad f(2, 0) = 4(2)^2 + 2(0)^3 = 16,$$

$$\text{Point D: } (-2, 0), \quad f(-2, 0) = 4(-2)^2 + 2(0)^3 = 16.$$

- **Subcase 2b:** $x_2 = \frac{4}{3}$

Substituting $x_2 = \frac{4}{3}$ into the constraint:

$$x_1^2 + \left(\frac{4}{3}\right)^2 = 4 \implies x_1^2 + \frac{16}{9} = \frac{36}{9} \implies x_1^2 = \frac{20}{9} \implies x_1 = \pm \frac{2\sqrt{5}}{3}.$$

This gives two points:

$$\text{Points E, F: } \left(\pm \frac{2\sqrt{5}}{3}, \frac{4}{3}\right).$$

Evaluating the function value:

$$f\left(\pm \frac{2\sqrt{5}}{3}, \frac{4}{3}\right) = 4\left(\frac{20}{9}\right) + 2\left(\frac{4}{3}\right)^3 = \frac{80}{9} + \frac{128}{27} = \frac{240 + 128}{27} = \frac{368}{27} \approx 13.63.$$

3. Summary of Critical Points and Conclusion

By comparing the values of all candidate points:

- The **minimum value** is -16 , at $(x_1, x_2) = (0, -2)$.
- The **maximum value** is 16 , at three points: $(0, 2)$, $(2, 0)$, and $(-2, 0)$.

Table 2: Evaluation of all critical points in S .

Location	Point (x_1, x_2)	λ	$f(x_1, x_2)$
Interior	$(0, 0)$	—	0
Boundary	$(0, -2)$	6	-16 (Global Minimum)
Boundary	$(0, 2)$	-6	16 (Global Maximum)
Boundary	$(2, 0)$	-4	16 (Global Maximum)
Boundary	$(-2, 0)$	-4	16 (Global Maximum)
Boundary	$\left(\frac{2\sqrt{5}}{3}, \frac{4}{3}\right)$	-4	$\frac{368}{27} \approx 13.63$
Boundary	$\left(-\frac{2\sqrt{5}}{3}, \frac{4}{3}\right)$	-4	$\frac{368}{27} \approx 13.63$

3 Problem 1.3

3.1 Part a)

Table 3: Genetic Algorithm Parameters

Parameter	Symbol	Value
Population size	N	100
Variable bound	a ($x_i \in [-a, a]$)	5
Number of genes	m	50
Number of variables	n	2
Tournament size	k	2
Tournament probability	p_{tour}	0.75
Crossover probability	p_c	0.8
Mutation probability	p_{mut}	0.02
Number of generations	G	2000

Table 4: Optimization Results across 10 Runs

Run	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{g}(\mathbf{x}_1, \mathbf{x}_2)$
1	2.999954	0.499989	3.363110×10^{-10}
2	3.000008	0.500002	1.060401×10^{-11}
3	3.000002	0.500001	7.924906×10^{-13}
4	2.999992	0.499998	9.710465×10^{-12}
5	2.999989	0.499997	1.828147×10^{-11}
6	2.999982	0.499996	4.960419×10^{-11}
7	3.000008	0.500002	1.003509×10^{-11}
8	3.000031	0.500008	1.516671×10^{-10}
9	3.000002	0.500001	7.924906×10^{-13}
10	2.999935	0.499984	6.836137×10^{-10}

3.2 Part b)

Table 5: Median Fitness Score for Varying Mutation Probabilities (100 Runs per Setting)

Mutation Probability (p_{mut})	Median Fitness
0.000	0.992701
0.005	0.997821
0.010	1.000000
0.020	1.000000
0.050	1.000000
0.100	0.999998

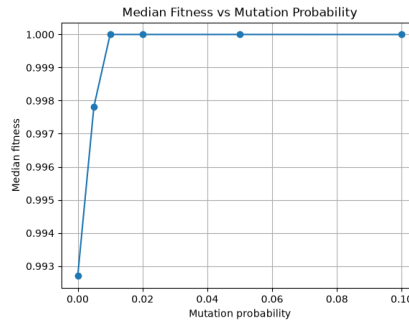


Figure 2: Median fitness vs Mutation Probability

Discussion on Optimal Mutation Probability:

Standard ($p_{\text{mut}} = 1/m$): For $m = 50$ genes, the standard rule is $p_{\text{mut}} =$

$1/m = 0.020$. On average one gene was mutated which in turn led to the optimal median fitness observed (1.000000).

Low Mutation ($p_{\text{mut}} \leq 0.005$): Too low of a mutation rate may lead to the algorithm not creating new genetic material and in turn getting stuck at points that are not the global minima (0.992701 at $p_{\text{mut}} = 0.000$).

High Mutation ($p_{\text{mut}} = 0.100$): A too high mutation rate can cause high-scoring chromosomes to be changed too much which in this case likely caused a lower median fitness score (0.999998).

Optimal Range: The optimal probability for mutation in this case is determined to be $p_{\text{mut}} \in [0.010, 0.050]$. The standard rule $p_{\text{mut}} = 1/m = 0.020$ and probabilities in its proximity is therefore regarded as the most optimal setting.

3.3 Part c)

The estimated guess based on the 10 runs in part a) is that the minimum is at $(x_1^*, x_2^*)^T = (3, 0.5)$.

Below is the analytical proof that $(x_1^*, x_2^*)^T = (3, 0.5)$ is a stationary point of the function:

$$g(x_1, x_2) = (1.5 - x_1 + x_1x_2)^2 + (2.25 - x_1 + x_1x_2^2)^2 + (2.625 - x_1 + x_1x_2^3)^2,$$

we want to show that the gradient is 0 at the point $(x_1^*, x_2^*)^T = (3, 0.5)$, $\nabla g(3, 0.5) = \mathbf{0}$.

Partial derivative with respect to x_1 : Differentiate with the chain rule:

$$\begin{aligned} \frac{\partial g}{\partial x_1} &= 2(1.5 - x_1 + x_1x_2) \frac{\partial}{\partial x_1}(1.5 - x_1 + x_1x_2) \\ &\quad + 2(2.25 - x_1 + x_1x_2^2) \frac{\partial}{\partial x_1}(2.25 - x_1 + x_1x_2^2) \\ &\quad + 2(2.625 - x_1 + x_1x_2^3) \frac{\partial}{\partial x_1}(2.625 - x_1 + x_1x_2^3) \\ &= 2(1.5 - x_1 + x_1x_2)(x_2 - 1) \\ &\quad + 2(2.25 - x_1 + x_1x_2^2)(x_2^2 - 1) \\ &\quad + 2(2.625 - x_1 + x_1x_2^3)(x_2^3 - 1) \end{aligned}$$

Partial derivative with respect to x_2 : Differentiate with the chain rule:

$$\begin{aligned} \frac{\partial g}{\partial x_2} &= 2(1.5 - x_1 + x_1x_2) \frac{\partial}{\partial x_2}(1.5 - x_1 + x_1x_2) \\ &\quad + 2(2.25 - x_1 + x_1x_2^2) \frac{\partial}{\partial x_2}(2.25 - x_1 + x_1x_2^2) \\ &\quad + 2(2.625 - x_1 + x_1x_2^3) \frac{\partial}{\partial x_2}(2.625 - x_1 + x_1x_2^3) \\ &= 2(1.5 - x_1 + x_1x_2)(x_1) \\ &\quad + 2(2.25 - x_1 + x_1x_2^2)(2x_1x_2) \\ &\quad + 2(2.625 - x_1 + x_1x_2^3)(3x_1x_2^2) \end{aligned}$$

Calculate gradient at $(x_1, x_2) = (3, 0.5)$: First evaluating the three base parenthetical terms:

$$\begin{aligned} 1.5 - 3 + 3(0.5) &= 1.5 - 3 + 1.5 = 0 \\ 2.25 - 3 + 3(0.5^2) &= 2.25 - 3 + 0.75 = 0 \\ 2.625 - 3 + 3(0.5^3) &= 2.625 - 3 + 0.375 = 0 \end{aligned}$$

All three terms equal 0, giving each term in both partial derivatives a multiple of 0:

$$\left. \frac{\partial g}{\partial x_1} \right|_{(3,0.5)} = 2(0)(0.5 - 1) + 2(0)(0.25 - 1) + 2(0)(0.125 - 1) = 0$$

$$\left. \frac{\partial g}{\partial x_2} \right|_{(3,0.5)} = 2(0)(3) + 2(0)(3) + 2(0)(2.25) = 0$$

Conclusion:

$$\nabla g(3, 0.5) = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

means that the point $(x_1^*, x_2^*) = (3, 0.5)$ is a stationary point.